

# Three Open Problems on Summing Multilinear Operators Resolved by Later Sharp Theorems

Literature-implied answer for arXiv:1204.5411

Model: GPT5.6

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**Status: literature-implied answer.** All three problems in Section 4 of Pellegrino's source paper are answered by combining its transfer formulas with the later optimality theorems [2, 3]. The deductions below are not presented as new theorems.

## 1 Source questions

The source paper [1, p. 8, Section 4] asks:

1. whether, for some  $\theta \in [0, 1]$ , the coincidence

$$\mathcal{L}({}^n\ell_1; \ell_2) = \Pi_{\text{as}}\left(\frac{2}{1+\theta}; \frac{2n}{n+\theta}, \dots, \frac{2n}{n+\theta}\right)({}^n\ell_1; \ell_2) \quad (1)$$

is sharp;

2. for real  $s > 2$  and integer  $n \geq s$ , the exact value of

$$\begin{aligned} M_{s,n} &:= \sup\{r : \mathcal{L}(E_1, \dots, E_n; \mathbb{K}) \\ &= \Pi_{\text{mult}}(2; r, \dots, r)(E_1, \dots, E_n; \mathbb{K}) \\ &\text{for all cotype-}s \text{ spaces } E_j\}; \end{aligned} \quad (2)$$

3. whether  $M_{s,n}$  can equal the analogous absolutely-summing supremum outside the case  $(s, n) = (2, 2)$ .

## 2 Problem 1: affirmative, for every $0 < \theta \leq 1$

Put  $p_\theta = 2/(1+\theta)$  and  $q_\theta = 2n/(n+\theta)$ . The cotype-2 transfer identity used in the source gives, for  $1 \leq t \leq 2$ ,

$$\Pi_{\text{as}}(p_\theta; t, \dots, t)({}^n\ell_1; \ell_2) = \Pi_{\text{as}}(\rho_\theta(t); 1, \dots, 1)({}^n\ell_1; \ell_2), \quad \rho_\theta(t) = \frac{p_\theta t}{np_\theta(t-1) + t}. \quad (3)$$

Here  $\rho_\theta(q_\theta) = 2/(n+1)$  and  $\rho_\theta$  is strictly decreasing. Pellegrino and Seoane-Sepúlveda proved that  $2/(n+1)$  is the optimal output exponent in  $\mathcal{L}({}^n\ell_1; \ell_2) = \Pi_{\text{as}}(r; 1, \dots, 1)$  [2, Theorem 1.1].

If the common input exponent in (1) could be increased from  $q_\theta$  to some  $t > q_\theta$ , take  $t$  sufficiently close to  $q_\theta$  that  $t \leq 2$ . Equation (3) would give coincidence at  $\rho_\theta(t) < 2/(n+1)$ , contradicting that theorem. The same calculation, with the output exponent decreased slightly from  $p_\theta$ , again produces an exponent below  $2/(n+1)$ . Thus (1) is sharp in both natural exponent directions for every  $0 < \theta \leq 1$ . In particular, Problem 1 has an affirmative answer. No claim about the separate endpoint  $\theta = 0$  is needed for the source question.

### 3 Problems 2 and 3: exact values

Araújo and Pellegrino prove that every scalar  $n$ -linear form on arbitrary Banach spaces is multiple  $(2; 2n/(2n-1), \dots, 2n/(2n-1))$ -summing [3, Theorem 3.2(i)]. Hence

$$M_{s,n} \geq \frac{2n}{2n-1}.$$

For the reverse inequality, specialize their Corollary 3.1 to multilinearity  $n$ , output exponent 2, and domain exponent  $p = s$ . Its hypothesis is  $2 \leq s < 2n$ , which follows from  $2 < s \leq n$ , and it gives

$$\sup\{r : \mathcal{L}({}^n\ell_s; \mathbb{K}) = \Pi_{\text{mult}}(2; r, \dots, r)({}^n\ell_s; \mathbb{K})\} \leq \frac{2n}{2n-1}.$$

Since  $\ell_s$  has cotype  $s$ , it is an admissible test space in (2). Therefore

$$\boxed{M_{s,n} = \frac{2n}{2n-1}}, \tag{4}$$

and the supremum is attained. This answers Problem 2.

The source itself proves that the corresponding absolutely-summing supremum is attained and equals

$$A_{s,n} = \frac{2sn}{2sn + s - 2n} \quad [1, \text{Eq. (2.10)}].$$

Combining this with (4), direct cross-multiplication gives

$$M_{s,n} = A_{s,n} \iff n = s.$$

Thus Problem 3 has a complete classification: equality holds for every integer pair  $(s, n) = (k, k)$  with  $k > 2$ , and fails whenever  $n > s$ .

### 4 Provenance and scope

The supporting paper [3] explicitly cites page 194 of the published source and improves the interval containing  $M_{s,n}$ , but it does not state the  $\ell_s$  specialization above or the equality classification. The paper [2] proves the endpoint optimality used for Problem 1 while identifying a different named conjecture. Exact searches did not locate a paper stating the three deductions together. Accordingly this packet records a *literature-implied answer*, not a new proof result.

### References

- [1] D. Pellegrino, *Sharp coincidences for absolutely summing multilinear operators*, Linear Algebra Appl. 440 (2014), 188–196; arXiv:1204.5411.
- [2] D. Pellegrino and J. B. Seoane-Sepúlveda, *Grothendieck’s theorem for absolutely summing multilinear operators is optimal*, Linear Multilinear Algebra 63 (2015), 554–558; arXiv:1307.4809.
- [3] G. Araújo and D. Pellegrino, *Optimal estimates for summing multilinear operators*, Linear Multilinear Algebra 65 (2017), 930–942; arXiv:1403.6064.